

Semi-Automatic Construction of Virtual Reality Environment for Highway Work Zone Training using Open-Source Tools

Can Li*

Zhu Qing†

Praveen Edara‡

Carlos Sun§

Bimal
Balakrishnan¶
Mississippi State
University

Yi Shang±

University of Missouri

University of Missouri

University of Missouri

University of Missouri

Mississippi State
University

University of Missouri

ABSTRACT

In this paper, we present a computer vision-based approach to assist in creating a virtual reality (VR) environment for training state transportation agency staff that inspect highway work zones. Current VR work zone environments are manually created by developers based on drive-through videos or work zone plans, which require a considerable amount of time. We leveraged two open-source tools, YOLO and Road Architect in Unity, to improve the efficiency of environment creation. The YOLO open-source tool processes the video input and creates an output file that contains work zone features (i.e., roadway geometrics and traffic signs) and their locations. The virtual environment in Unity was created by Road Architect using its user-friendly interface. Our results showed that the proposed method was able to significantly reduce the development time for creating training modules. The semi-automatic method can be applied to other training applications where VR-environment creation is time consuming.

Index Terms: Computing methodologies - Artificial intelligence - Computer vision - Computer vision problems - Object detection; Applied computing - Operations research – Transportation

1 INTRODUCTION

Work zones present challenges to drivers, non-motorists, and highway workers. Drivers experience lane reductions, narrow lanes and shoulders, reduced speed limits, presence of workers and construction equipment, and temporary traffic control devices in work zones. These atypical driving conditions require drivers to pay increased attention and can increase driver stress. Non-motorized users such as pedestrians, wheelchair users, and bicyclists also face challenges in work zones. These challenges stem from constrained roadway geometrics such as narrow facilities, increased proximity to traffic and construction, temporary closures or detours of sidewalks/pathways, maintenance of accessibility standards on those detours, and lack of signage to accommodate these modes. Risk to construction and maintenance workers stems from two sources – 1) working next to live traffic and 2) working with and around heavy construction equipment and hazardous site conditions. Given the safety implications, state transportation agencies must regularly inspect work zones for compliance. Recently, virtual reality (VR) has been incorporated into work zone training modules [1][2]. Due to the diversity and complexity of work zones, current VR training modules can only cover limited scenarios, and more realistic work zone environments are needed to be developed. However, developing realistic work

zone environments requires considerable time to understand work zone features and manually create 3D environments. In order to enhance the efficiency of work zone reconstruction in VR, we leveraged two open-source tools (YOLO and Road Architect) to design a new semi-automatic method to create a virtual work zone in Unity based on a drive-through video.

YOLO is a state-of-the-art open-source neural network for real-time object detection. There are a few different versions of pre-trained YOLO models available on GitHub. They can be finetuned on domain-specific data for work zone traffic sign detection [3]. Road Architect is an open-source tool developed by MicroGSD for Unity [4], which provides a user-friendly interface to create roads, intersections, bridges, traffic signs, and other road objects. Road Architect uses a parametric approach to simplify the roadway environment design.

In this paper, a video of a work zone recorded from a camera mounted on the windshield of a vehicle driving through a work zone was used to demonstrate the proposed method.

2 RELATED WORK

Automated virtual environment creation simplifies the workflow and saves time in the development of VR applications such as digital twins and immersive simulations [5]. The automated methods mainly include image-based, laser scanning-based, and radio-based approaches, digital modeling, and manual methods [6]. The image-based construction approaches can either use matching features to generate the semi-dense/dense scene [7][8] or use image processing algorithms to detect geometric primitives and key features from images [9], relying on a large number of images or videos as input. The laser-scanning approach is considered a top-prioritized method due to its geometrical accuracy [10][11]. However, laser scanning is also costly and inconvenient with a high possibility of data loss [12]. The radio-based methods with a long-read range can also track and collect dynamic components and material information. However, the work of installation, scanning, and maintenance of radio-based methods requires trained workers and is labor-intensive. Digital modeling methods such as GIS and BIM offer a parametric modeling approach to support virtual environment development [13][14]. While these promising automatic and semi-automatic methods have been implemented in other disciplines, existing VR applications for work zones still rely on manual development to suit specific tasks such as testing work zone safety devices [15] and evaluating work zone setups [16]. Computer vision has been widely used in transportation from assisting autonomous vehicles in analyzing their surroundings [17] to evaluating traffic sign readability [18]. For detecting traffic signs, computer vision applications can be categorized as traditional machine learning-based and deep learning-based methods [19]. For the machine learning-based methods, the two main structures are based on AdaBoost and Support Vector

* e-mail: lican@mail.missouri.edu

† e-mail: zqing@missouri.edu

‡ e-mail: edarap@missouri.edu

§ e-mail: csun@missouri.edu

¶ e-mail: bbalakrishnan@caad.msstate.edu

± e-mail: shangy@missouri.edu

Machine (SVM) [22][23]. For example, AdaBoost was used to detect work zone traffic signs for reliable autonomous driving [17], and SVM was applied for highway work zone recognition [22]. For deep learning-based methods, convolutional neural network (CNN) based models have achieved state-of-the-art performance for traffic sign detection. For example, the YOLO model has been applied for real-time traffic sign detection [24], while R-CNN has been proven in large-scale traffic sign detection [24].

3 DATASET

While YOLO has been used for traffic sign detection, its application specifically to detect work zone signs is limited. Thus, we collected work zone sign data to train a YOLO object detection model. We used a camera to record 83 minutes of work zone videos. Those videos are taken at different times of the day (daytime and evening) and in various weather conditions (sunny, cloudy, and rainy) to make our object detection model more robust. Images containing work zone traffic signs were extracted from the recorded videos. These images were supplemented by images obtained from online searches. A dataset consisting of 200 images of four work zone traffic signs (i.e., ROAD WORK AHEAD, SPEED LIMIT 60, and two graphical signs of lane shift) was created. Figure 1 shows examples of the work zone traffic signs with different backgrounds. A 5:1 sample ratio was used to extract negative samples that did not contain any work zone traffic signs from the videos. For the labeling process, we use a tool called Roboflow to generate the bounding boxes for the work zone traffic signs for training the object detection model.



Figure 1: Examples of work zone sign dataset

4 SEMI-AUTOMATIC ENVIRONMENT RECONSTRUCTION

As shown in Figure 2, the semi-automatic environment reconstruction method includes four steps after obtaining a work zone video with GPS records. The first step is retrieving the roadway geometry information from the camera's GPS record and detecting work zone signs in the video. The second and third steps are converting the video and GPS output files to a single input file for Unity. The last step is the virtual reconstruction of the roadway and work zone signs. More details will be introduced in the following subsections.

4.1 GPS and Sign Detection

4.1.1 GPS

Work zone drive-through videos were recorded using an Olympus TG-Tracker, which saves its GPS location every 30 seconds. As the video recorder was mounted on the front windshield and no lane change was conducted in work zones, the GPS record collected the roadway geometric information and can be used to map the road

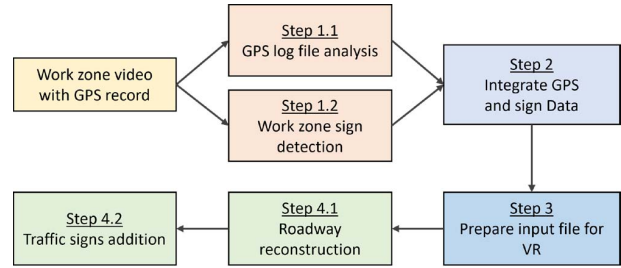


Figure 2: Virtual work zone reconstruction flow chart

alignment. In Step 1.1, the GPS record with timestamp is obtained as a text log file from the camera.

4.1.2 Sign Detection

In Step 1.2, work zone traffic signs in the drive-through video are detected using YOLO. Since the input of YOLOv5 requires images of 640×640 pixels, all images were converted into 640×640 pixels. The dataset was split into 80% for training, 10% for validation, and 10% for testing. A pre-trained YOLOv5 model was finetuned on the training dataset for 100 epochs with a batch size of 16. The detection results were promising with 91% mean average precision (mAP). Figure 3 shows examples of the detection results on the testing dataset. The results showed that YOLO quickly detected work zone signs.



Figure 3: Detection results

(Ground truth - top left: road work ahead; top right: two lane shift right; bottom left: speed limit 60; bottom right: two lane shift left)

4.2 Integrate GPS and Sign Data

In Step 2, the GPS data from Step 1.1 and sign information extracted in Step 1.2 are integrated into a single output file using a Python script. As shown in Table 1, the output file includes timestamp, GPS location (latitude and longitude), and type of work zone sign detected. The structure of the output file is relatively simple and detected signs can be easily verified.

Table 1: Example of the output file generated in Step 2

Timestamp	Latitude	Longitude	Work Zone Sign
5:09:12 PM	38°56'19.85"N	92°37'43.35"W	
5:09:42 PM	38°56'20.65"N	92°37'10.34"W	
5:10:12 PM	38°56'21.01"N	92°36'36.81"W	
5:10:23 PM			ROAD WORK AHEAD
5:10:42 PM	38°56'21.89"N	92°36'3.65"W	
5:11:12 PM	38°56'22.57"N	92°35'31.65"W	
.....			

4.3 Prepare Input File for Virtual Reality

In Step 3, an input file is generated for recreating the VR environment using the open-source tool, Road Architect in Unity. During the conversion, GPS records were translated to road nodes in Unity that were then used to build the road layout. The work zone sign data was converted to its model name in Road Architect. Since GPS data is collected only at 30-second intervals, the location of a sign needs to be estimated using the timestamps. For example, if a ROAD WORK AHEAD sign is recorded at 5:10:27 PM in the video and the locations of the before and after road nodes were saved at 5:10:12 PM and 5:10:42 PM, the sign's location is estimated as the midpoint of the two road nodes. The structure of the new input file is shown in Table 2.

Table 2: Example of the input file for VR

Node	Position.X	Position.Z	Code	Distance
1	0	0		
2	6.578	-1008.639		
3	9.538	-2033.167		
Sign			GSDFedSign-988- GSDFedSign-988- RoadWorkAhead	506.611
4	10.196	-3046.389		
5	12.827	-4024.167		
.....				

4.4 Virtual Reality Environment Generation

The environment generation can be divided into two sub-steps: Step 4.1, the creation of the roadway and Step 4.2, the addition of work zone signs. The roadway can be created by entering the road nodes data in the input file (Table 2) into Road Architect. After the roadway was created, work zone signs were added at a specific distance from the previous road node, as described in the previous section. Road Architect provides a graphic user interface to guide users to manually add a sign. The task of adding work zone signs could also be completed through a custom C# script in Unity. Based on the input file shown in Table 2, a sign of the same model name was added at 506.6 m after the third road node. A screenshot of the

Road Architect user interface to add the road nodes and work zone signs is shown in Figure 4.

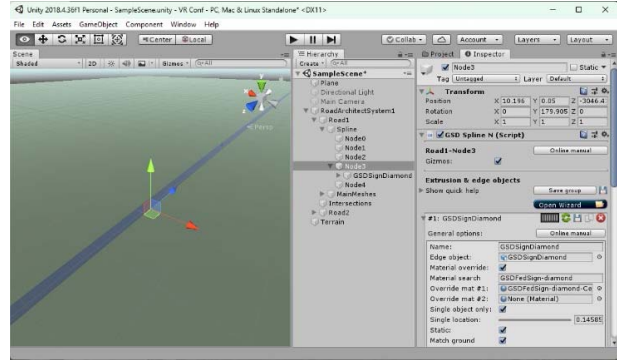


Figure 4: Road nodes and work zone signs in Road Architect

5 RESULTS

With the use of two open-source tools, we presented a semi-automatic method to reconstruct a virtual work zone using real-world video data. The development time was significantly reduced by using open-source tools to assist in analyzing the real-world environment and facilitating parametric design. For example, in a recent driving simulator-based study [26], a virtual work zone similar to the one in the drive-through video was constructed in three days relying on the work zone plan and satellite imagery. The semi-automatic method proposed in this paper completed the same scenario in less than one day. Implementing the new method required developing new scripts from scratch to help users format the input and output files between software, as the two open-source tools were not designed for collaborative use. Open-source tools with input/output APIs are able to facilitate further redevelopment and collaborative use.

The quality of the VR environment constructed by the new method was evaluated through a frame-by-frame comparison with the original video (Figure 5). The comparison showed that the layout of the virtual roadway was close to the real roadway, although there were slight differences in curve radius, which can be improved by using an extra GPS tracker with a higher update rate. Regarding the locations of work zone signs, the estimated errors were about ± 90 feet. Since the sign spacing is from 1,000 to 1,500 ft in a work zone with a speed limit of 60 mph, such errors were considered acceptable.

Overall, the semi-automatic approach with the use of two open-source tools allows new users to quickly create VR environments and offers easier scalability.



Figure 5: Work zone signs and roadway comparison

6 LIMITATIONS AND FUTURE WORK

The training dataset used in the current study is relatively small and only focuses on four work zone traffic signs. Next steps include collecting additional data consisting of other common traffic signs under various roadway and weather conditions. Such a dataset will help build a more robust algorithm to detect work zone signs from video. Future research could also explore collecting and using terrain data to help improve the VR environment. Finally, the proposed approach can be tested for other traffic management applications such as incident management training.

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